

only supports 20MHz channels that can carry a maximum of 54 Mbps of raw data per channel. 802.11n increases that throughput to 150 Mbps per channel. It also employs a technique called channel bonding to combine two adjacent 20MHz channels into a single 40MHz channel. The technique more than doubles the channel bandwidth to 300Mbps throughput. Channel bonding is most effective in 5GHz frequency given the far greater number of available channels, while 2.4GHz frequency has only three non-overlapping 20MHz channels.

Packet aggregation. Packet aggregation increases efficiency by aggregating multiple packets of application data into a single transmission frame. 802.11n networks can send multiple data packets with the fixed overhead cost of just a single frame. Packet aggregation is more beneficial for data applications such as file transfers. However, real-time applications, e.g., voice, do not benefit from packet aggregation. For voice, it is better to minimise the number of 'packed' packets to reduce latency and eliminate jitter contentions.

Practical considerations for deploying 802.11n

Before deploying 802.11n technology, enterprises need to take into account certain basic issues like power over Ethernet, traffic forwarding and current LAN architecture. Here is a

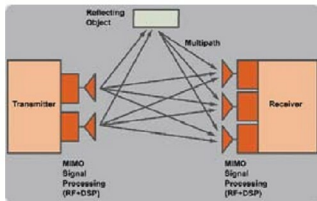


Fig. 1: MIMO and multipath effect

checklist for network administrators who are planning to deploy 802.11n technology.

Power over Ethernet. Power over Ethernet (PoE) is a technology used on wired Ethernet LANs. It is defined by the IEEE 802.3af specification and allows the electrical current necessary for the operation of a remote device to be carried over the Ethernet LAN cables rather than by power cords. The IEEE 802.3af standard states exactly how much power is to be provided over the Ethernet cable.

Given the advanced capabilities of 802.11n access points—MIMO, SDM and multiple antennae—it is easy to see that the components may be very power hungry. This is especially true when running in full 3x3 MIMO mode, or when using two radios to support simultaneous 2.4GHz and 5GHz operation. In such a situation, standard PoE switches (IEEE 802.3af) are not capable of feeding enough power to these '11n' access points.

Most vendors provide special power injectors along with 802.11n access points that can feed the required additional power (typically 18W to 20W per access point). Alternatively, some vendors reduce the performance of the access point, thereby reducing the power consumption so that it can be connected to the existing PoE switch.

Forwarding of wireless traffic. When planning for 802.11n, consider how much traffic your wireless network will generate on your wired LAN. Many enterprise WLAN architec-

tures require access points to backhaul all traffic to a centralised controller, which then routes it across the wired network.

Multiple 802.11n access points can produce a much higher load on your wired LAN than legacy 802.11a/b/g access points. This depends on the size of the WLAN deployment and on how much applica-

tion data is forwarded to the controller. Therefore distributing or bridging traffic at the access point should be seriously considered instead of tunneling all traffic centrally.

Wired LAN architecture. The traditional 802.11a/b/g access points give a raw data rate of max. 54 Mbps, hence the current wired LAN structure of 100Mbps connectivity to each access point is sufficient. The new 802.11n access points provide a data rate of 150-300 Mbps. In this situation, 100Mbps wired network is not sufficient. For getting the required 150-300Mbps throughput from the wireless network, the wired LAN should also be upgraded to Gigabit connectivity to access points.

Recommendations for IT managers

The time is now. Consider 802.11n wireless deployment in your enterprise if:

1. You are in the process of evaluating for green-field deployment
2. You are upgrading the wireless infrastructure from standalone access points to a controller-based system
3. There are reliability and throughput issues with your existing WLAN

Dual-radios are best. It is highly recommended to deploy concurrent 2.4GHz and 5GHz radio coverage while deploying '11n,' where the access point supports both 2.4GHz and 5GHz 802.11n simultaneously. Chances are that all the wireless clients will not be upgraded to 802.11n

TABLE II
Various Classes of PoE as Defined by IEEE 802.3af

Class	Usage	Maximum power level at the input of powered device (watts)
0	Default	0.44 to 12.94
1	Optional	0.44 to 3.84
2	Optional	3.84 to 6.49
3	Optional	6.49 to 12.95
4	Reserved	(PSEs classify as Class 0)